

SmartDRM: Blockchain based Digital Rights Management System

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Problem Statement

- ▶ Centralized control leading to single-point failure
- ▶ Lack of transparent ownership verification
- ▶ Manual and unreliable license enforcement
- ▶ No intelligent mechanism to detect abnormal usage patterns
- ▶ Limited auditability and accountability

Objective

- ▶ To design a decentralized DRM framework using blockchain technology
- ▶ To ensure immutable ownership and license records through smart contracts
- ▶ To provide controlled and flexible access mechanisms for digital assets
- ▶ To detect suspicious usage behavior using AI-based analysis
- ▶ To maintain an auditable and transparent activity log
- ▶ To develop a modern, user-friendly interface for asset management

Existing System

- ▶ Current DRM solutions primarily operate using centralized servers managed by content distributors or platform providers.
- ▶ Characteristics:
- ▶ Central authority controls access and licenses
- ▶ User authentication and permissions stored centrally
- ▶ DRM policies enforced at application level

Disadvantage

- ▶ Single-point failure and high vulnerability to attacks
- ▶ Limited transparency in ownership verification
- ▶ Dependency on trusted third parties
- ▶ Difficult to audit access and usage
- ▶ Poor scalability for large digital ecosystems

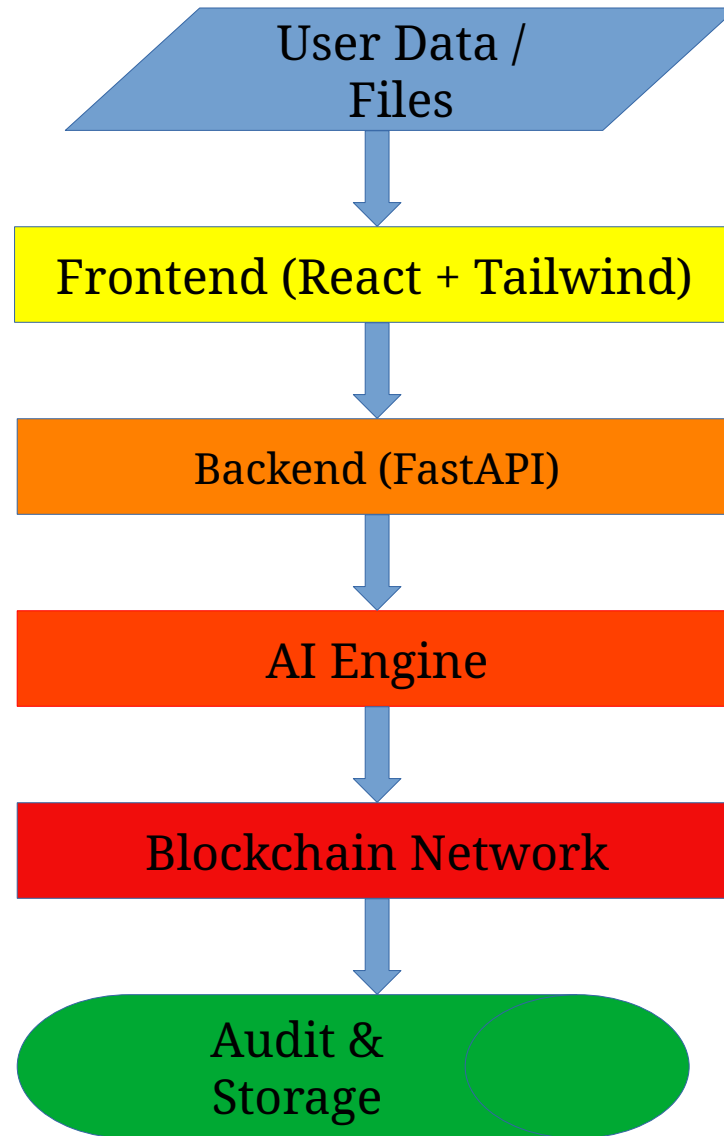
Proposed System :

- ▶ Blockchain-based asset registration and ownership proof
- ▶ Smart contract-driven license issuance and enforcement
- ▶ AI-powered behavioral risk analysis
- ▶ Secure audit logging for all critical actions
- ▶ Full-stack web interface for real-time interaction

Advantage

- ▶ Immutable and tamper-proof ownership records
- ▶ Transparent license enforcement
- ▶ Reduced piracy and unauthorized access
- ▶ Automated royalty-ready transaction tracking

SYSTEM ARCHITECTURE/FLOW DIAGRAM



Modules :

- ▶ Authentication Module
- ▶ Asset Management Module
- ▶ Blockchain & Smart Contract Module
- ▶ License & Access Control Module
- ▶ AI-Based Risk Analysis Module
- ▶ Audit Logging Module
- ▶ Frontend Dashboard Module

MODULE 1: Authentication Module

- ▶ Purpose: Secure onboarding and identity verification of all users accessing protected digital assets.
- ▶ Key Features: Email/password login, secure password hashing, and session/token-based authentication.
- ▶ Role-Based Access: Differentiates admins, content owners, and consumers with tailored permissions.
- ▶ Security Controls: Brute-force protection, validation of credentials, and secure storage of user data.
- ▶ Integration Points: Acts as the entry gate to all other modules (assets, blockchain, AI risk, and dashboard).

MODULE 2: Asset Management

- ▶ Purpose: End-to-end lifecycle management of protected digital assets.
- ▶ Asset Types: Supports documents, media, and other licensed digital content.
- ▶ Responsibilities: Upload, encrypt, store, organize, and distribute assets.
- ▶ Metadata Management: Tags, categories, owners, and sensitivity levels.
- ▶ Goal: Central, secure repository to manage all DRM-protected content

Architecture & Storage :

- ▶ Architecture: API layer + storage backend with encryption at rest.
- ▶ Upload Pipeline: Validation, virus/malware checks (optional), encryption, and indexing.
- ▶ Storage Model: Encrypted blobs with separate metadata records in the database.
- ▶ Linking: Assets are linked to owners, licenses, and blockchain records.
- ▶ Scalability: Designed to handle large numbers of assets and high request volumes.

Operations & Benefits:

- ▶ Core Operations: Upload, update, retire, and retrieve assets with proper permissions.
- ▶ Search & Filter: Metadata-based search by owner, tag, date, and license state.
- ▶ Security Controls: No direct file URLs; all access goes through policy checks.
- ▶ Monitoring: Asset usage statistics feed into risk analysis and dashboards.
- ▶ Benefits: Reduced leakage risk, better asset visibility, and streamlined content operations.

Module 3: Blockchain & Smart Contract

- ▶ Purpose: Provide immutable, verifiable records for licenses and key asset events.
- ▶ Smart Contracts: Encode usage terms (duration, region, limits) as on-chain logic.
- ▶ Ledger Role: Acts as the “source of truth” for rights and critical transactions.
- ▶ Trust Model: Reduces reliance on a single central authority.
- ▶ Goal: Enhance transparency, non-repudiation, and auditability of DRM policies.

Design & Components

- ▶ On-Chain Data: License identifiers, ownership transfers, and important asset events.
- ▶ Off-Chain Links: References to encrypted assets and local database records.
- ▶ Contract Functions: Issue, renew, revoke, and verify license conditions.
- ▶ Security:
Carefully designed contract interfaces to prevent misuse and re-entrancy issues.
- ▶ Integration: Backend interacts with blockchain node / provider for transactions and queries.

Workflows & Benefits

- ▶ License Issuance Flow: Admin creates license → contract call → on-chain record.
- ▶ Verification Flow: Access control module queries contract state before granting access.
- ▶ Audit Flow: Investigators can replay on-chain history to validate claims.
- ▶ Interoperability: Potential to integrate with external marketplaces or rights systems.
- ▶ Benefits: Strong evidence trail, reduced disputes, and higher stakeholder confidence.

Conclusion

- ▶ SmartDRM-X demonstrates how blockchain and artificial intelligence can be effectively integrated to build a secure, transparent, and intelligent Digital Rights Management system. By decentralizing ownership records, automating license enforcement, and introducing behavioral analysis, the system significantly improves trust and security in digital content distribution.
- ▶ The proposed framework is scalable, extensible, and suitable for applications in education, digital media platforms, enterprise document management, and software licensing.
- ▶ Future enhancements may include integration with decentralized storage systems and advanced machine learning models for predictive piracy detection.



THANK YOU !